

GRASSHOPPERS AND CRICKETS

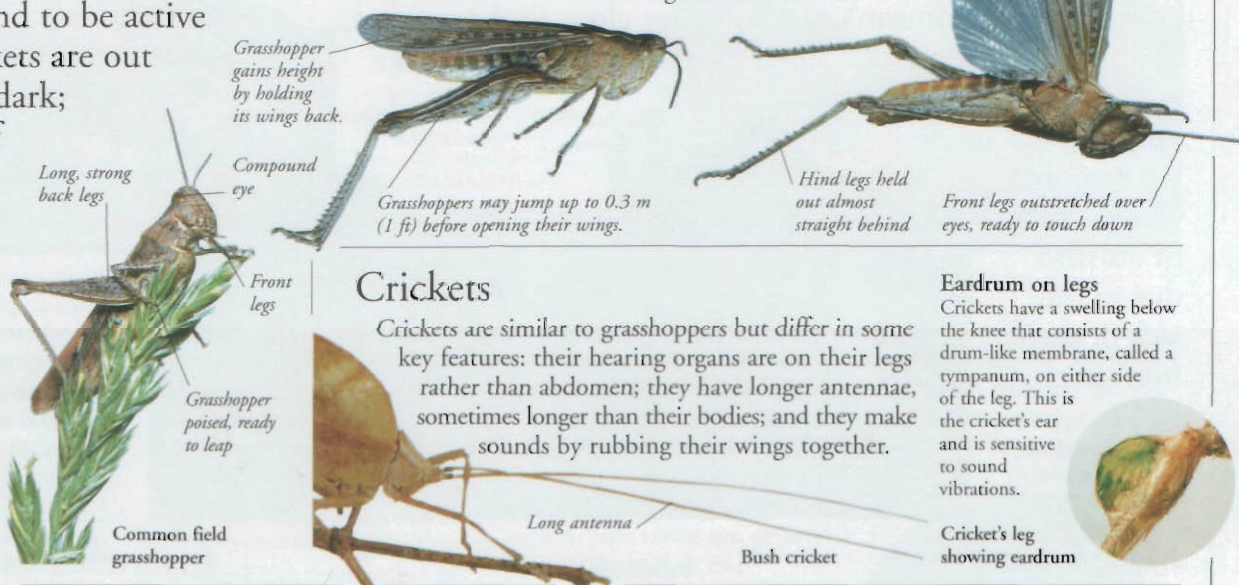


FAMOUS FOR THEIR ATHLETIC LEAPS and chirping calls, grasshoppers and crickets are among the largest and most distinctive of insects. Most are weak fliers and prefer to move by walking or jumping. They live mainly in grasslands and rainforests, but some live in deserts and caves, and a few wingless species burrow underground.

Grasshoppers tend to be active by day, but crickets are out and about after dark; in many parts of the world their constant chirps fill the night air.

Leaping

Propelled forward by snapping their hind legs straight, grasshoppers can out-jump all other insects. If danger threatens or if the grasshopper wants to move to another clump of vegetation, it springs into the air, opens and flutters its wings to prolong the leap, and drops down as much as 1 m (3.3 ft) ahead.



Crickets

Crickets are similar to grasshoppers but differ in some key features: their hearing organs are on their legs rather than abdomen; they have longer antennae, sometimes longer than their bodies; and they make sounds by rubbing their wings together.

Eardrum on legs

Crickets have a swelling below the knee that consists of a drum-like membrane, called a tympanum, on either side of the leg. This is the cricket's ear and is sensitive to sound vibrations.

Cricket's leg showing eardrum

Locust swarms

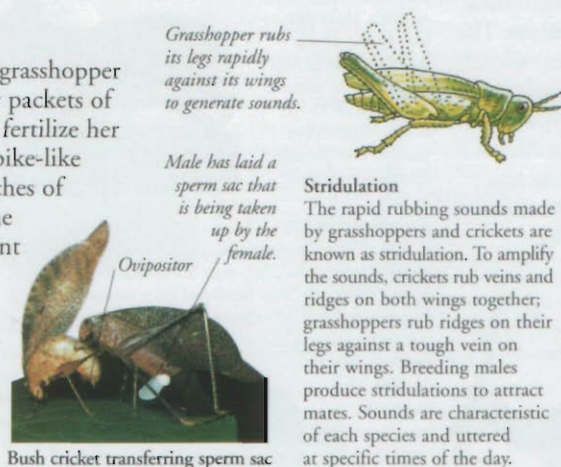
Locusts are grasshoppers. After heavy rains, lush plant life grows, creating the right conditions for locusts to breed in large numbers. Swarms of up to 50 billion set out across the land. They devastate crops and plants, causing famine.



Swarm of locusts in Ethiopia

Reproduction

During mating a male grasshopper or cricket transfers tiny packets of sperm to the female to fertilize her eggs. She then uses a spike-like ovipositor to place batches of up to 100 eggs at a time into the soil or into plant stems and leaves. Tiny nymphs – miniature versions of the parents – hatch from the eggs. They moult and grow many times until they reach adult size.



Feeding

Most grasshoppers feed on leaves, buds, and other parts of plants that they chew with their mouthparts. Crickets have a more varied diet. Many eat plants but also catch and devour other insects – in fact, some bush crickets are dedicated hunters. Crickets that live in houses and caves scavenge on dead and waste matter.



Bush cricket eating a grasshopper

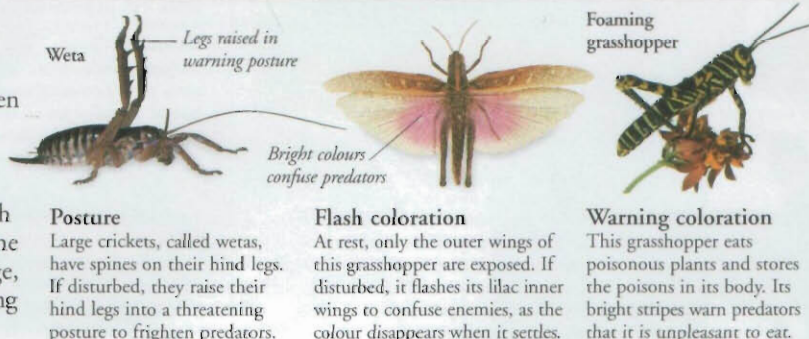
Great green bush cricket

Stridulation

The rapid rubbing sounds made by grasshoppers and crickets are known as stridulation. To amplify the sounds, crickets rub veins and ridges on both wings together; grasshoppers rub ridges on their legs against a tough vein on their wings. Breeding males produce stridulations to attract mates. Sounds are characteristic of each species and uttered at specific times of the day.

Defence

Many grasshoppers and crickets are brown or green so they are less visible to predators. Others have brightly coloured hind wings that they flash to warn off enemies. Some have elaborate camouflage, with body parts resembling leaves and plant stems.



COMMON FIELD GRASSHOPPER

SCIENTIFIC NAME *Chorthippus brunneus*

ORDER Orthoptera

FAMILY Acrididae

DISTRIBUTION Europe

HABITAT Dry open areas with short grass

DIET Grass and other low-lying plants

SIZE Length: males up to 18 mm (0.7 in); females up to 23 mm (0.9 in)

LIFESPAN 6–7 months

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CAVE WILDLIFE

CAMOUFLAGE AND COLOUR

GRASSLAND WILDLIFE

INSECTS

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